

Shlok Shivaram Iyer

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A motivated Computer Science student(2026 batch), passionate about building new solutions and learning new technologies.

EDUCATION

B.M.S. College of Engineering

Pursuing B.E in Computer Science

Nov 2022 - Present

CGPA: 8.8

Narayana PU college JP Nagar

Senior Secondary Education

Jun 2020 - Apr 2022

2nd PUC Percentage: 93.16%

Clarence Public School

High School Education (ICSE)

June 2008 - March 2020

X ICSE Percentage: 90%

EXPERIENCE

Project-based Learning Intern, Sigma-Aldrich

July 2025 – Aug 2025

Technologies: Gemini, Firebase, Model Fine-tuning, Firebase Cloud Functions

- Developed Sahayak, a web app designed to help teachers explain concepts with local analogies.
- Developed using HTML, CSS, JavaScript, Python and Firebase for the backend.
- Used Gemini's multimodal capabilities. Fine-tuned a Llama 3-8B Instruct model on an Indian history and geography dataset to help generate local analogies.
- All backend functions were deployed as Firebase Cloud Functions. Output supported in regional languages to help teachers teach in rural areas.
- [Github link](#)

PROJECTS

Enhancing Long Document Summarization | *Ongoing project*

Mar 2025 - Present

- Analysing different Extractive and Abstractive summarization techniques to improve long document summarization.
- Developing a new model using **Hierarchical Attention Mechanism** and **Graph Neural Networks**.
- Implements a discourse-aware summarization framework with the HAN capturing the hierarchy of the document and the GNN focusing on the facts.
- Using libraries like transformers, AllenNLP, torch, sqLite, etc
- Est completion Nov/Dec 2025

Waste Detection and Classification | *Object Detection - YOLO v5s, ConvNeXT*

Nov 2024 – Jan 2025

- **YOLOv5s model** for object detection to detect waste in public spaces and **ConvNeXT** fine-grained classifier to classify them into sub-categories.
- Detects 7 main classes of waste along with 28 sub-classes; trained on over **30,000 images**.
- Used a combination of YOLOv5s and ConvNeXT along with Stratified sampling of the dataset for better training accuracy
- Achieved an mAP of **81.7%** and an F1 score of **81.3%** for the YOLOv5s object detection model and ConvNeXT classifier achieved a test accuracy of **90.43%** and an F1 score of **92.93%**
- Simple UI interface created with Streamlit UI to test the model
- [Github link](#)

- **Mobile application for project management** specifically for software projects. Implemented using Figma for initial design and Flutterflow for application development.
- Features include dynamic progress tracking with graphs and progress bars, task management via custom labels, user-friendly timeline and an interactive profile page.
- [Github link](#)

TECHNICAL SKILLS

Languages : Python, Java, C/C++,
Frameworks : Flask, Flutter, Hadoop, Apache Spark, Hugging Face
Libraries : Pandas, NumPy, Matplotlib, Scikit-learn
Developer Tools : GitHub, VS Code, Flutterflow
Databases : MySQL, MongoDB, Firebase, Apache Cassandra
Certification : [Coursera Machine Learning Specialization](#)

STUDENT VOLUNTEERING

Content team member | at *CODEIO BMSCE*

Jan 2024 - Aug 2024

- Content writer for the CODEIO club at BMSCE.
- Supported event coordination by creating reports and written content, ensuring effective documentation and communication of club initiatives.